



Research Article

Characterizing Future Trends of Temperature and Rainfall Using CMIP6 Climate Scenarios over Upper Wabe-Sheble River Basin, Ethiopia

Dejen Terefe Cherinet* 

Meteorological Weather Forecast, Ethiopia Meteorology Institute, Robe, Ethiopia

Abstract

Climate change poses significant challenges on agriculture water resources and energy sectors mainly induced by human activities. Climate models' scenarios play vital role in assessing future temperature and rainfall their impacts on various sectors. This study aims to assessed future rainfall and temperature trend Using CMIP6 under the SSP2-4.5 and SSP5-8.5 scenarios over Wabe-Sheble river basin. Data Eight GCMs under were obtained from ESGF, whereas observed data from Ethiopia meteorology institute. An Ensemble model was Best performed for precipitation (RMSE=4.6, NSE=0, $r=0.9$) and maximum temperature (best performed for minimum temperature relative to individual models. Linear scaling bias correction method was applied to reduced model errors and improve repetitiveness for local climate features. During the near-term period (2031–2060), future rainfall is projected to increase by 17.0%, 7.4%, and 12.9% under SSP2-4.5, while by 22.7%, 15.3%, and 17.4% increases under SSP5-8.5 scenarios during Belg, Kermit, and annual respectively). During the far term (2061–2090) mean maximum temperature ($^{\circ}\text{C}$) is projected to increase under the SSP2-4.5 by 1.2 $^{\circ}\text{C}$, 1.6 $^{\circ}\text{C}$, 1.8 $^{\circ}\text{C}$, 1.6 $^{\circ}\text{C}$, whereas, under increase by 3.2 $^{\circ}\text{C}$, 3.7 $^{\circ}\text{C}$, 3.9 $^{\circ}\text{C}$ and 3.8 $^{\circ}\text{C}$ Bega, Belg Kermit and annual periods respective. Under the SSP2-4.5 scenario future mean minimum temperature ($^{\circ}\text{C}$) is projected increases /relative by 2.1 $^{\circ}\text{C}$, 2.2 $^{\circ}\text{C}$, 3.1 $^{\circ}\text{C}$ and 3.2 $^{\circ}\text{C}$ during near-term (2031–2060, whereas during the far term (2061–2090) increase 3.0 $^{\circ}\text{C}$, 2.5 $^{\circ}\text{C}$, 3.1 $^{\circ}\text{C}$ and 3.1 $^{\circ}\text{C}$ Bega, Belg Kermit and annual period. Over all, this finding reveals both temperature and precipitation is projected to increases under the SSP2-4.5 and SSP5-8.5 scenarios. Although, future rainfall is projected to increase temperature also increase simultaneously therefor future warming expected to affects agriculture, water resource, range land ecosystems, therefore timely climate information and interventions activities are recommended. Further studies are suggested considers drought and flood assessment and their potential changes under future climate scenarios over the study area.

Keywords

Climate Change, CMIP6, Climate Scenarios, Trend Analysis, Future Climate

*Correspondence: Dejen Terefe Cherinet (dejmet2012@gmail.com)

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1. Introduction

Climate change deviates weather pattern from the long-term mean value for extended period, mainly driven by anthropogenic activities associated with green house emission. Global atmospheric temperature rises 1.1 °C period 2011-2020 relative the base line 1850-1990 associated with human induced greenhouse gases emission [1]. Robust currently warming driven buy also exhibited over Africa [2]. Seasonal annual based Tmax and Tmin experienced increases trend across Ethiopia [3]. Spatial and temporal rainfall variability was experienced at global scale resulting occurrence of wet and dry events [4]. Significant spatial temporal rainfall variability affects water resource sectors over Africa [5]. Stream flow and perception increases in the wet season while decrease in dry season associated with projected climate change [6]. Future rainfall and temperature projection under the CMIP6 scenario experienced increasing trend high and low emission scenarios [7]. Observed change in weather and climate extremes affects ecosystems global to regional scale [8].

significantly impact the agricultural sector at different depending on ago ecological zone [9]. Projected future climate change induced drought frequency and intensity events resulting challenges agriculture yield reduction over Africa [10]. The impact is huge over sub-Saharan Africa over the community their economic sectors rely on pastoral economic sectors [11]. Consider major play a major role on Agriculture production back bone GDP of a country [12]. Follows crop yield reduction at present and future climate scenario associated Kermit and Bega temperature increment [13]. Climate change induced temperature and rainfall variability highly effect on cereal crop production [14]. Climate change alters normal river flow pattern, subsequently leads low flow and high flow extremes. Future hydrological drought across the globe with global hydrological model forced by GCM RCP8.5 high flow increase for northern part while low flow over spot areas arrived by S. Shrestha et al., (2021). Low flow, stream flow and precipitation are increased whereas severe drought decreased over upper Blue Nile river basin [15]. According to [16] rainfall, maximum and minimum temperature increase both SSP2-4.5 and SSP5-8.5 but anomalous warming leads to more evaporation, altering freshwater availability. Reduce national GDP about 8-10% moreover resulting 2.4 million people under food insecurity level by 50 [17]. Belg and Kermit rainfall exhibited significant variation the leads socioeconomic sectors of the area [18]. Future climate change highly influence watershed ecosystem, therefore understanding proactively vital for early warning planning and climate adaptation activities. For assessing future climate change and its impacts across different sectors under various socioeconomic pathways, the Coupled Model Intercomparison Project Phase 6 (CMIP6) provides a paramount contribution.[19]. CMIP6 models were also applied for projecting future temperature and rainfall associated climate change over Africa [20]. CMIP6 under the

(SSP2-4.5) and SSP5-8.5) scenarios was applied to assess future climate trend and extreme precipitation associated with climate change across Ethiopia [21]. Main source of developing early warning, adaptation planning at different spatial scale to tackle future climate change impacts [22]. Previous study of future trend of temperature and rainfall using CMIP5 under the RCP4.5 and RCP8.5 over Wabe-shebele basin [23]. Recently future climate change assessment but using limited stations and GCMs models under the SSP2-4.5 and SSP5-8.5 scenarios up to 2085 [7]. Unlike previous assessments based on a limited number of GCMs and meteorological stations, this study employs eight GCMs gives more comprehensive assessment of future temperature and rainfall changes. The analysis also considers seasonal and annual timescales and extends the projection period to 2100, This provides valuable information for early warning, climate adaptation planning activities at local area level.

2. Materials and Methods

2.1. Description of the Study Area

The study was conducted in the upper portion of Wabe-Shebele river basin located, Northern and highland parts main basin. Geographically the study extends from latitude of 6°46'.55" to 8°54' 05"N 39°25'.06"- 40°34'42"E incorporates Arsi, West Arsi and some parts of Bale zones (*Figure 1*). Total geographical area covers about s varying topographic features range between 460 to 4200m elevation range (m.a.s.l) above sea level. Major land use land cover composition classes are rangeland, Crops, Trees (47.6%), (30.2%), 19.9 (%) (*Figure 2*) respectively.

2.2. Climate Characteristics

The Upper Wabe-shebele river basin characterized by bimodal rainfall regime Kermit extends from (June-September) whereas Belg (March-May) months. The highest amount of monthly rainfall was received on August 174.5 (mm) during Kermit season, whereas on April 116.0 (mm) during Belg season. in contrast, lowest amount of rainfall is received on the month December and January (*Figure 3*). The highest maximum mean temperature is recorded in March (24.0 °C), the lowest mean is in August (21.5 °C), whereas the lowest mean minimum temperature recorded in January (5.3 °C), whereas the highest August (8.5 °C) (*Figure 3*). The study area seasonal rainfall patterns influencing of by large-scale atmospheric and ocean systems including ENSO, IOD, ITCZ, and subtropical westerly jet (STWJ) and Arabian High [24-26].

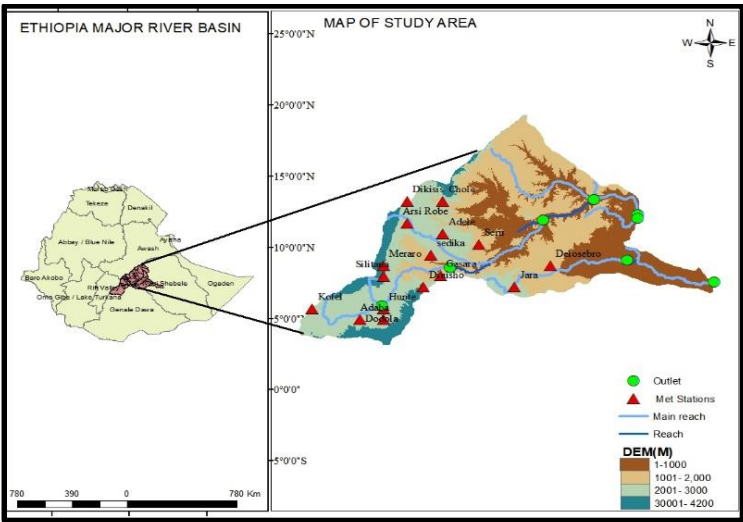


Figure 1. Location of the Study Area.

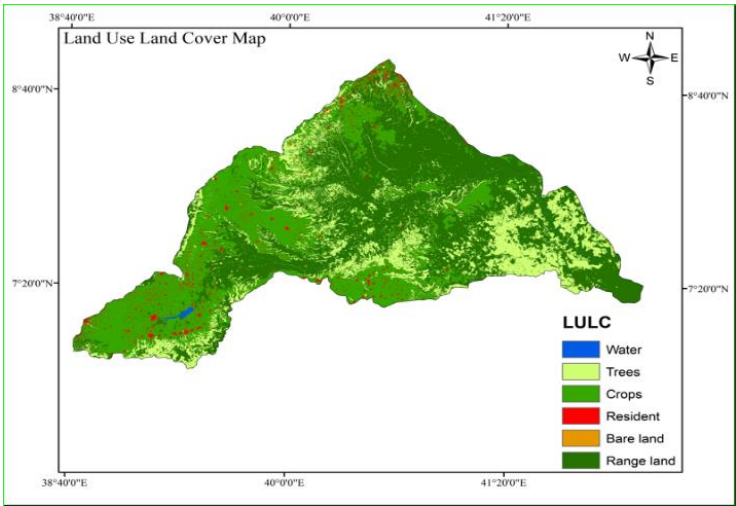


Figure 2. Land use land Cover Map.

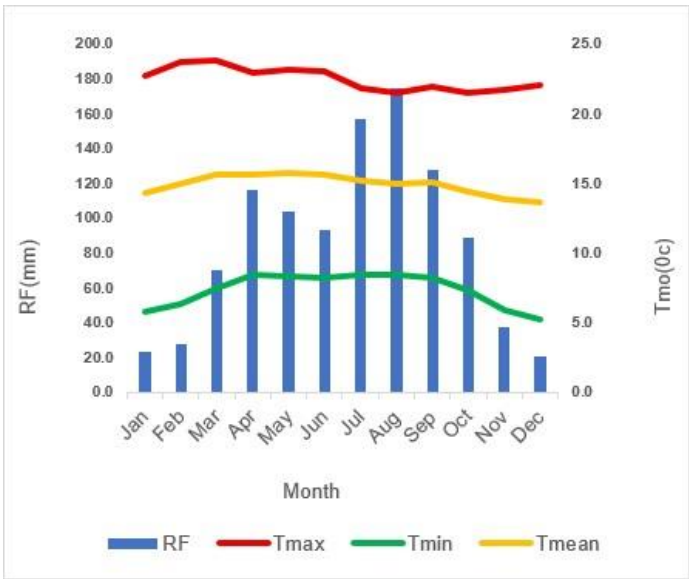


Figure 3. Monthly Climatology Rainfall, Tmax, Tmin and Tmean of the study Area.

2.3. Data Collection

2.3.1. Historical Observed Data

Daily observed meteorological data, including rainfall, maximum temperature, and minimum temperature, for the period 1991–2020 were collected from the Ethiopian Meteorology Institute (EMI).

2.3.2. GCMs Data Collection

Daily rainfall, maximum and minimum temperature of eight Global climate model (GCM) Climate model intercomparison phase six (CMIP6). For this study GCM under the SSP2-4.5 and SSP5-8.5 scenarios were obtained from world climate research Group (WCRP) earth science grid federation (ESGF).

2.4. Data Preprocessing

2.4.1. Data Quality Control

Collected rainfall data was captured in Microsoft excel 2016, random visualization recoded calendar format.

Make sure rainfall recorded value shouldn't be negative and converted Trace to 0.0 [27].

Missis data was identified and gabs completion treated with high spatial and temporal resolution grid data (4*4) km. Grid dataset generated with combines interpolated in situ measurements from over 500 NMA rain gauges with satellite estimates (Dinku et al. 2014).

Outlier data values were detected with using the Turkey fence approach The rules of this approach are that inner fence are located at a distance 1.5 times interquartile range below the lower and above the upper quartiles and outer fences are located a distance 3 times the interquartile range below the lower and above the upper quartiles. Values outside the turkey fences are identified [28].

Equation of turkey fence

$$[Q_1 - 1.5 * IQR, Q_3 + 1.5 * IQR]$$

Q_1 = first quartile

Q_3 = third quartile

2.4.2. CMIP6/GCMs Data Preprocessing

For projecting future potential change of climate its impact on different sectors, state of the art CMIP6 model was applied. The SSP2-4.5 and SSP5-8.5 scenarios were employed, the study period classified near-term (2031-2060), far-term (2061-2090). Collected GCMs data preprocess were carried out Using Climate data operator (CDO) tool. Merging time, unit conversion, set standard calendar, and extraction with longitude and latitude with study area data preprocess steps were undergone. Climate model has different spatial resolution; the model was remapped using bilinear method common resolution to (4*4) km ENACT data. Climate models have different resolution there for to evaluate them should be re-mapping to common resolution [29]. Climate models were selected based on their ability to reproduce the historical climate conditions of the study area. The performance of the individual GCMs was evaluated using statistical performance metrics, including the correlation coefficient (r), root mean square error (RMSE), and Nash–Sutcliffe efficiency (NSE). These metrics were used to assess the agreement between the simulated and observed climate variables, and the best-performing models were selected for future climate projections. The performance of the selected GCMs were evaluated using coefficient of correlation (r), Root Mean Square Error (RMSE), and Nash Sutcliff Error (NSE) statistical metrics. Climate models appropriate for future climate change study were selected with similar way arrived by [30, 31].

Climate models may have limitation of capturing local-scale climate features appropriately sometimes under estimate and other times over estimate observation data. To minimize model uncertainty errors linear-scaling bias correction technique was applied Downscaling is based on the average difference between monthly observed time series and historical run time series of GCM/RCM over same period of the observed series additive correction is preferable for temperature whereas multiplicative correction is preferable to variables like precipitation, vapor pressure, solar radiation etc [32].

Table 1. GCM Performance Evaluation statistical Metrics.

Model Evaluation Statistical metrics	Description	Range	Perfect square
$(r) = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$	Coefficient of correlation	-1 to 1	1
$RMSE = \sqrt{\frac{\sum_{i=1}^n (X-Y)^2}{n}}$	Root means square error	0 to ∞	0
$NSE = 1 - \frac{\sum_{i=1}^n (X - \bar{Y})^2}{\sum_{i=1}^n (X_{his} - \bar{X}_{his})^2}$	Nash-Sutcliffe Coefficient	$-\infty$ to 1	1

2.4.3. Trend Analysis of Future Rainfall

The Mann Kendall (MK) is the non-parametric time series trends test less affected with outlier's value (Mann, 1945; Kendall, 1975). Applied to study hydrological time series and climatic variability (Longboard and Villani, 2010). Mostly used to detect an upward and downward trend i. e., monotonic trends in a series of environmental, climate, and hydrologic series of data. Particularly the null hypothesis considers no trend whereas the alternative hypothesis shows the trend in the two-sided test one-sided may upward trend or downward trend [33]. Sen's slope (SS) estimator is another non-parametric technique mostly detects the magnitude the trend datasets. This technique is used for the identification of trend magnitude.

$$S = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \text{sgn}(X_j - X_i)$$

where S was the Mann-Kendal's test statistics x_i and x_j were the sequential data values of the time series in the years i and j ($j > i$) and N was the length of the time series. Positive S value indicates an increasing trend and a negative value indicates a decreasing trend in the data series. The sign function was computed as

The sign function computes as below

$$\text{sgn}(x_j - x_i) = \begin{cases} 1 & \text{if } (x_j - x_i) > 0 \\ 0 & \text{if } (x_j - x_i) = 0 \\ -1 & \text{if } (x_j - x_i) < 0 \end{cases}$$

The variance of S, for the situation where there may be ties (that is, equal values) in the x values is given by:

$$\text{Var}(s) = \frac{1}{8} [N(N-1)(2N+5) - \sum_{i=1}^m t_i(t_i-1)(2t_i+5)]$$

where, m is the number of tied groups in the data set and ties the number of data points in the i th tied group.

$$\text{ZMK} = \begin{cases} \frac{s-1}{\sqrt{\text{var}(s)}} & \text{if } s > 0 \\ 0 & \text{if } s = 0 \\ \frac{s+1}{\sqrt{\text{var}(s)}} & \text{if } s < 0 \end{cases}$$

The presence of a statistically significant trend was evaluated using the ZMK value. In a two-sided test for trend, the null hypothesis H_0 was accepted if $|ZMK| < Z_{1-\alpha/2}$ at a given level of significance. $Z_{1-\alpha/2}$ was the critical value of ZMK from the standard normal table. For example: for 5% significance level, the value of $Z_{1-\alpha/2}$ is 1.96. positive value of ZMK indicates an increasing trend while a negative value indicates a decreasing trend. In the present study, the significance of observed change is examined at $p \leq 5\%$ significance level confidence interval. Therefore taking 0.05 significance level as reference, the Sen's slope estimator was computed as:

$$T_i = \frac{x_i - x_k}{j - k} \text{ for } i = 1, 2, 3, \dots, N$$

$$Q_i = \begin{cases} T \frac{T+1}{2} & \dots \dots \dots N = \text{odd} \\ \frac{1}{2} \left\{ T \frac{N}{2} + T \frac{N+2}{2} \right\} & \dots \dots \dots \text{even} \end{cases}$$

Sen's estimator was computed as $Q_{\text{med}} = T(N+1)/2$ if N appears odd and it is considered as $Q_{\text{med}} = (TN/2 + T(N+2)/2)$ if N appears even. At the end, Q_{med} was computed by a two-sided test at 100 (1- α) % confidence interval and then a true slope was obtained by the non-parametric test. Positive value of Q indicates an upward or increasing trend and negative value shows decreasing trend and if the values are zero, it shows the data are fluctuate around theme. Positive value of Q_i indicates an upward or increasing trend and a negative value indicates downward or decreasing trend in the time series. All the tests were measured at significant level (0.05) or 95% confidence of interval. Sen's slope magnitude negative indicated decreasing value and positive magnitude increasing value P value $\leq \alpha$ value test has statistically significant trend while P value $> \alpha$ value has statistically non-significant trend. Mann Kendall trend test carried out over stations at seasonal and annual time. R-Studio on window environment that is supported by R software with R4.3.3 and Modified package engaged for computation purpose. From Mann Kendall trend test increasing and decreasing trend statistically significantly or none significantly identified. The trends in projected future rainfall and temperature were analyzed at seasonal and annual time scales for the (2031–2090) periods under the SSP2-4.5 and SSP5-8.5 scenarios.

2.4.4. Change in Future Rainfall

Change of future average rainfall in (%) relative the baseline period (1991-2020) was calculated by the difference between mean future rainfall and baseline rainfall value divided by baseline mean rainfall multiplied by 100. Baseline period was (1991-2020) whereas future period near term (2031-2060) and far-term (2061-2090). Change of future rainfall in (%) was computed both SSP2-4.5 and SSP5-8.5 climate scenarios with both periods, Positive values indicate an increase in rainfall relative to the baseline period, whereas negative values indicate a decrease.

$$\text{Percent of change of rainfall} = \left(\frac{\text{Future} - \text{Baseline}}{\text{Baseline}} \right) * 100$$

2.4.5. Change in Future T(max)

The Change in future average maximum temperature (Tmax) under the climate scenarios b relative to baseline

(1991-2020) was computed by subtracting future mean temperature climate scenario under the SSP2-4.5 and SSP5-8.5 near-term (2031-2060) and far-term (2061-2090) periods. The

negative value indicates decreasing whereas positive value increasing future maximum temperature respect to baseline periods.

$$\text{Change of future } T(\max) = (\text{Future Mean } T(\max) - \text{Baseline mean } T(\max))$$

2.4.6. Change in Future T(min)

The Change in future average minimum temperature (Tmin) under the climate scenarios b relative to baseline (1991-2020)

$$\text{Change of future mean } T(\min) = \text{Future mean } T(\min) - \text{Baseline mean } T(\min)$$

was computed by subtracting future mean temperature climate scenario under the SSP2-4.5 and SSP5-8.5 near-term (2031-2060) and far-term (2061-2090) periods. The negative value indicates decreasing whereas positive value increasing future maximum temperature respect to baseline periods.

time.

3. Results and Discussions

The model evaluation statistics indicated that multi-model Ensemble (MME) of Eight GCMs was selected with respect to individual models. Therefore, climate change assessment was done with multi model ensemble (MME) rainfall and maximum temperature data under the SSP2-4.5 and SSP5-8.5 scenarios. The use of Multimodal Ensemble of GCMs best representation future climate impact assessment than individual models [34-37]. According to [38] the assessment of future temperature and rainfall over Mediterranean region Ensemble of six GCMs was exhibited better than individual models based performance evaluation metrics. An ensemble of six GCMs was selected as best-performed for future precipitation assessment under the SSP2-4.5 and SSP5-8 scenarios [39] (Therefore, the multi-model ensemble (MME) was found to perform better than individual models. Moreover, combining multiple GCMs helps reduce the uncertainty associated with individual models and provides a more robust representation of future climate conditions. Among eight GCMs, CNAR-CM6-1 (Centre National de Recherche Météorologiques France) model was selected using model performance evaluation statistics future scenario minimum temperature assessment under climate scenarios (*Error! Reference source not found.*). CNAR-CM6-1 (Centre National de Recherche Météorologiques model was selected for climate change study presented by [40]. Performance evaluation of multiple GCMs showed that CNRM-CM6-1 was among the best-performing models for minimum temperature at both daily and monthly timescales over Ethiopia [41]. Identifying the best-performing GCM was not sufficient, since GCMs have inherent uncertainties and errors in representing local-scale climate features. Therefore, bias correction was undertaken to reduce model uncertainty and systematic biases and to improve the reliability of GCM outputs for future climate impact assessment [42]. Bias corrected GCM output showed closer agreement with observation value compared to uncorrected output, for climate assessment application [43]. Downscaling is based on the average difference between monthly observed time series and historical run time series of GCM/RCM over same period of

3.1. Trend Analysis of Future Rainfall

The Mann Kendall trend test was conducted using bias corrected rainfall data under the SSP2-4.5 and SSP5-8.5 scenarios. Trend tests were measured at significant level 0.05 or (95% confidence of interval) for both seasonal and annual period. Trend of test of rainfall under SSP2-4.5 scenario showed significantly increases about (7.2-16.3), (4.6-16.5), (9.8-26.3) and (29.4 53.1) mm/decade for Bega, Belg, Kermit and annual respectively, across most stations.

(Table 2). This result is consistent with previous finding on trend test of rainfall under the SSP2-4.5 scenario, precipitation showed increasing trend about 7.77% and 13.74% at the annual and seasonal time scale respectively [44]. Under the SSP2-4.5 and SSP5-8.5 scenario run off was projected increase this linked with increasing future precipitation arrived by [45] under the SSP2-4.5 and SSP5-8.5 scenarios future temperature and precipitation showed increasing trend relative to the baseline period [21]. Recent study reported by [32] reported that future seasonal and annual projected to increase significantly under the SSP2-4.5 and SSP5-8.5 scenarios seasonal rainfall at the range of (9-73%) .

According to [46]. Future Precipitation projected to increase about 0.42 to 2.82 mm mid-century 0.15 to 3.79 by the end of century per year under the SSP2-4.5 and SSP5-8.5 scenarios. Frequency of future hydro climate extremes is mainly association with projected increase precipitation about (35%), (43%) and (38%) in annual, Spring and short (OND) time period respectively across east Africa presented by [47]. The trend test under SSP2-4.5 scenario future precipitation non-significantly decreases with 2.0 and 0.5 mm/decade only at Dodola and Jara stations in Belg season. Previous study by [48] also revealed Belg rainfall statistical non-significant decreasing trend at 1.93 mm/decade. Trend analysis Belg rainfall over Ethiopia did not show significant trend reported by [49]. Under the SSP2-4.5 scenario trend of rainfall slightly decreases but under SSP5-8.5 increased on the mid-century [50]. The BCC-CSM-2MR model precipitation projection revealed none significant trend under the SSP scenarios at the long-term period [51]. Kermit rainfall indicated decreasing trend at

15.23% under SSP2-4.5 scenario for the period 2071-2100 [52]. Under the SSP5-8.5 scenario, future precipitation showed significantly increasing trend about at (12.6-32.4), (7.9-16.9), (11.0-31.2) and (23.8-71.0) mm/decade during Bega, Belg, Kermit and annual period respectively, across most stations (Table 3). [53] reported that future precipitation projections showed an increasing trend across most stations during the 2050s (2041–2060) and 2080s (2071–2090). Under the SSP5-8.5 scenario Belg rainfall was projected non-significantly increase across Seru, Chole, Adaba, Delosebro and Jara stations. Under the SSP2-4.5 and SSP5-8.5 climate scenarios annual rainfall showed non-significantly increasing trend during mid-term and far-term at (2.0–11.9) (6.1–16.1 (mm/decade) [54].

Under the warmest (SSP5-8.5) scenario annual rainfall pattern altered, moreover increases extremes JJAS rainfall about (25-30)% [55]. Future rainfall increases Under the ssp1-2.6, SSP2-4.5 and SSP5-8.5 scenarios elevate runoff [56]. Overall, man Kendall trend test of future precipitation predominantly showed increasing under both SSP2-4.5 and SSP5-8.5 scenarios on contrast some stations was experienced non-significant trends.

3.2. Projected Changes in Future Rainfall Under CMIP6 Scenarios

The Change of mean future rainfall (%) relative to baseline period was computed with SSP2-4.5 and SSP5-8.5 climate scenarios at midterm (2031-2060) and far term (2060-2090). Under the SSP2-4.5 climate scenario future the average rainfall increases on near term (2031-2060) by (17%), (7.4%) and (12.9%) in and, during far-term (2061-2090) increases with (18.3%), (15.6%) and (20.2)% Belg, Kermit and annual periods across nine stations respectively (Table 4). Under the SSP2-4.5 scenario future monthly rainfall is projected to increases from 0.42 to 2.82% in mid-century and 0.15 to 3.79% the far century whereas under the SSP5-8.5 scenario increases midcentury 1.45 to 5.51% and far-term 2.57 to 9.8% by the end of far century upper awash river basin [57]. Previous study reported that future projected rainfall increase streamflow at 13–17% [58]. Under the SSP2-4.5 and SSP5-8.5 scenarios annual mean rainfall projected to increase during 2050 and 2080 period over Pravara river basin in India [59]. Under SSP2-4.5 scenario future rainfall decreases about (11.9%) over Arsi robe Jara Silitan and Dinisho Belg season (Table 4). Annual mean rainfall patten showed decreasing pattern under both SSP2-4.5 and SSP5-8.5 scenarios presented by [60]. Rainfall showed decreasing by 15.23% under the SSP2-4.5 during 2071-2100 Kermit season [61]. Kermit rainfall show increasing by +44.28% and +59.34%, whereas Bega rainfall showed decreasing over upper Wabe below bridges under both RCP4.5 and RCP 8.5 scenarios [62]. Future climate change under SSP1-2.6 SSP2-4.5 and SSP5-8.5 scenarios revealed annual precipitation and temperature increases during spring and winter whereas, decrease during summer and autumn over

arid region [63]. The change in (%) future mean under the SSP5-8.5 scenario showed increasing during near future (2031-2060) with (22.7%), (15.3%) and (17.4%) Belg, Kermit and annual respectively, whereas far-term (2060-2090) increases (31.6%), (24.7%) and (30.6%), Belg, Kermit and annual period Future annual rainfall projected increases under the SSP5-8.5, and SSP2-4.5, scenarios at 10.66% and 15.80%. [64]. Future precipitation is projected slightly increased at 13.7%, and 17.7% under SSP2-4.5 and SSP58 respectively reported by [65]. Future rainfall under the SSP5-8.5 scenario expected to increases about 45.41%, 149.40%, 52.26%, and 45.92% in R10MM, R20MM, RX1DAY, and RX5DAY associated with climate change [66] Future climate change under SSP2-4.5 scenario on stream flow water resources over the Upper Indus Basin of Pakistan precipitation increases about 13–17% presented by [67]. Future precipitation changes increases with $7 \pm 5\%$, $10 \pm 6\%$ and $\pm 13\%$ under SSP126, SSP2-4.5, and SSP5-8.5 in 2071–2100 compared to 1985–2014, respectively arrived by [68]. Kermit rainfall projected to decrease by 5.4% across Kofele, Hunte, Jara and Meraro stations whereas Belg rainfall decrease about 9.4 across Adele and silitana stations. Mean annual rainfall SSP1-2.6 and SSP5-8.5 scenarios is projected to decreases at 4.3% to 13.8% in northern part, increase at 14.5% to 25.4% in southeast semi-arid catchment over Tanzania [69]. Future SSP5-8.5 (warming scenario) supposed increases ocean temperature trigger tremendous evaporation, resulting more moisture-laden air when, converging the storm system can, generate robust precipitation [70]. Overall, under both SSP2-4.5 and SSP5-8.5 scenarios future precipitation is projected predominantly increase; this may be associated future anticipated warming which enhance evaporation increases atmospheric moisture availability this follows, however negative change also projected across some stations.

3.3. Projected Change in Future Maximum Temperature

The Change of future mean maximum temperature (°C) from baseline period under the SSP2-4.5 scenario is projected to increase during near-term (2031-2060) about 0.9, 1.0, 1.1 and 1.0) °C whereas during the far-term (2061-2090), increases at 1.2, 1.6, 1.8, 1.6 °C Bega, Belg Kermit and annual period respectively (Table 5). This is findings consists with previous study [71] using multi models ensemble under SSP1-2.6, SSP2-4.5, SSP5-8.5 scenarios projected to increase by 1.31 °C, 1.32 °C, 1.45 °C during near-term (2021–2040), 1.75 °C, 2.06 °C, 2.66 °C during the mid-term (2041–2060), and 1.08 °C, 2.97 °C, 5.62 °C in the long-term (2081–2100) period. Maximum temperature is projected increases at 0.8 °C and 2.8 °C under SSP2-4.5 and SSP5-8.5 climate scenario reported by [72]. This study consistence with previous finding that maximum temperature increases 0.92 °C and 1.86 °C during the 2040s and 2080s, under the SSP2-4.5 Over Genale dawa [44] Future temperature change under cmip6 scenario indicated increases at

3 °C and 1–4 °C under SSP2–4.5 and SSP5–8.5 scenarios respectively [73]. Annual mean maximum temperature show increment (1.3–2.0) °C for SSP2–4.5 and (1.7–2.3) °C for SSP5–8.5 in 2040–2069, at the same time, increments of (1.7–2.3) and (2.8–3.2) °C were predicted for 2070–2099 reported by [74]. Under the SSP2–4.5 scenario future maximum temperature is projected decreases during Bega with 0.4–2.6 °C over Dinisho and Seru stations season respectively, Belg decreases about 0.4–2.8°C, 0.3–0.5 °C and 0.2–1.0°C over Dinisho, Hunte and Seru stations respectively. Future decrease maximum temperature projected s under the SSP2–4.5 scenario over central Oromia [75]. The Change in future maximum temperature (°C) relative to the baseline period under SSP5–8.5 is projected to increases on near term (2031–2060) at 2.6, 2.3, 2.5 and 2.4 °C Bega, Belg Kermit and annual period respectively, whereas during far-term (2061–2090) 3.2, 3.7, 3.9 and (3.8) °C during Bega, Belg Kermit and annual period. Future temperature with the period (2050–2079) is projected to increases with 0.8 and 3.3 compared to baseline period [76]. Mean annul maximum temperature increases with near-far term period show increases of (1.5) °C, (2.2) °C, (2.8) °C, and (3.8) °C under the four SSP scenario respectively [77]. under the SSP5–5.8 scenario mean, maximum, and minimum temperature increases 2–3 °C, and 5–6 °C at the middle (end) of the current century over Europe [78]. Over east Africa future temperature and evapotranspiration increases associated with climate change [68]. Under the SSP5–8.5 scenario maximum temperature increases 80.5% and 4.8% at N45.5 °–60 °respectively [69]. Under SSP5–8.5 on Belg maximum temperature increase with (0.3–0.5) °C Over Dinisho, Hunte and Seru stations. Previous study reported under SP1–2.6, SSP2–4.5, and SSP5–8.5 scenarios annual mean temperature over East Africa increases relative to long term period about 1.4 °C, 2.3 °C, and 4.4 °C respectively [20]. regional climate change over east Africa revealed annual average temperature increases 2 °C toward midcentury (2021–2050) under SSP5–8.5 climate scenario linked with sever climate extremes reported [70]. Future CMIP6 climate change scenario showed both 1901–1940 (warming), 1941–1970 (cooling) and 1971–2014 (rapid warming) by [71]. Overall, under the SSP2–4.5 and SSP5–8.5 scenarios, future maximum temperature is projected to *predominantly increase relative to the baseline period across most stations*, which is associated with increased greenhouse-gas concentrations that enhance the *atmospheric trapping of outgoing terrestrial radiation*, thereby increasing radiative forcing and warming the near-surface atmosphere.

3.4. Projected Change in Future Minimum Temperature

The Change of future mean minimum temperature (°C) under both SSP2–4.5 and SSP5–8.5 scenarios relative to the baseline period. Under the SSP2–4.5 scenario future changes of mean minimum temperature projected increases relative to the baseline period at 2.1°C, 2.2 °C, 3.1°C and 3.2 °C during near-term (2031–2060) and far-term 3.0°C, 2.5°C, 3.1°C and

3.1°C Bega, Belg, Kermit and annual period respectively (Table 6). Climate change study under cmip6 of SSP5–8.5 over awash river basin Ethiopia in the far future maximum temperature will be arrived around 41–42 °C whereas minimum temperature 24°C respectively presented by [72]. Future minimum temperature is projected to increases by 0.8, 1.5, and 2.0 °C under SSP2–4.5, while, 1.0, 2.2, and 3.6 °C under SSP5–8.5, during 2030s, 2050s, and 2080s period respectively [73]. maximum and minimum temperature are projected to increase over Yangtze River Basin during (2025–2044) period at 0.09 °C, 0.29 °C and 0.66 °C per decade under the SSP1–2.6, SSP2–4.5, and SSP5–8.5 scenarios respectively. [74] reported that surface temperature is projected to increase under the SSP2–4.5 scenario across northwestern and eastern North America and Eastern Europe, moreover greater warming projected under the SSP5–8.5 scenario. Under the SSP5–8.5 scenario, future mean minimum temperature is projected to increase relative to the baseline period by 0.3–4.3, 0.4–4.4, 1.0–5.9, and 1.7–5.2 °C during the near-term (2031–2060) and by 3.4–5.4, 3.5–5.7, 3.8–5.2, and 4.8–5.8 °C during the far-term (2061–2090) for the Bega, Belg, Kiremt, and annual periods, respectively. Minimum temperature change will increases of the period (2020–2049) and (2050–2079) respect to baseline period (1985–2014) increases by 0.8 and 3.3 °C [66]. Under ssp1–2.6 and SSP5–8.5 scenarios minimum temperature increases between 0.2 °C and 4.5 °C [72]. Under the SSP5–8.5 and SSP2–4.5 minimum temperature increases during winter time of by 5 °C in northern parts of India by the end of the 21st century [75]. Similarly, [76] projected that mean minimum temperature would increase by 1.17, 1.91, 2.80, and 3.69 °C under the SSP1–2.6, SSP2–4.5, SSP3–7.0, and SSP5–8.5 scenarios, respectively, during the far-future period (2075–2100). Under the SSP5–8.5 climate scenario *over tropics minimum temperature increase throughout the century particularly at southern Europe for the 2081–2100 [67]. Similarly, [77] projected that mean annual minimum temperature is projected to increase by 1.96 °C under SSP2–4.5 while by 3.11 °C under SSP5–8.5 during the 2031–2060 period relative to the 1985–2014 baseline. The findings is consistence with [35]. Mean minimum temperature rises about 1.83, 2.33 and 2.85 °C under SSP2–4.5, SSP3–7.0, and SSP5–8.5 scenarios respectively. Under high emission SSP5–8.5 scenario both maximum and minimum temperature are projected increase [78].

Subsequently under both SSP2–4.5 and SSP5–8.5 scenarios minimum temperature predominantly positive change relative the bassline period across most stations this is associated with human induced greenhouse gases concentration in the atmosphere traps emitted radiation from the earth.

4. Conclusion and Recommendations

The study assessed future change of temperature and precipitation using CMIP6/GCMs under SSP2–4.5 and SSP5–8.5 scenarios with study period near-term (2031–2060) and far-

term (2061-2090) period over wabe shebele basin. An Ensemble of eight GCMs was selected for future rainfall and maximum temperature assessment whereas, CNAR-CM6-1 (Centre National de Recherche Météorologiques France) was selected minimum temperature based on model performance evaluation metrics. Climate models have systematic biases, therefore, the selected models were bias-corrected using linear-scaling bias correction technique. Future rainfall is projected to increase across most station during Bega, Belg, Kermit and annual seasons period under both SSP2-4.5 and SSP5-8.5 scenarios while decreasing trend across limited stations. Future rainfall is projected to increase under both SSP2-4.5 and SSP5-8.5 scenarios, potentially due to enhanced evaporation, atmospheric moisture availability, and moisture transport under a warming climate, which contribute to increased precipitation. Under SSP5-8.5 and SSP2-4.5 scenarios future maximum temperature and minimum temperature predominantly projected to increase while, projected decreasing over limited

stations. The predominant future warming is primarily associated with increasing anthropogenic greenhouse-gas emissions, which enhance radiative forcing and lead to a rise in atmospheric and surface temperatures.

The finding indicated both temperature and precipitation are projected increases under the SSP2-4.5 and SSP5-8.5 scenarios. Although, future rainfall is projected to increase temperature also increase simultaneously under the SSP2-4.5 and SSP5-8.5 scenarios, therefore future moisture stress is expected due to atmospheric warming, therefore warmed atmosphere expected to affects agriculture, water resource, range land ecosystems. For agricultural sector select drought tolerate and short cycle crop varieties, harvesting rainwater, waters and natural resources management. Policymakers and decision makers and private investors consider future anticipation of climate change before implementing far-reaching development project and investment over the study area. Future studies consider drought and flood assessment and their potential changes under future climate scenarios over the study area.

Abbreviations

BCC-CSM2-MR	Beijing Climate Center Climate System Model, Medium Resolution
CDO	Climate Data Operators
CMIP5	Coupled Model Intercomparison Project Phase 5
CMIP6	Coupled Model Intercomparison Project Phase 6
CNRM-CM6-1	Centre National de Recherches Météorologiques Climate Model Version 6-1
EMI	Ethiopian Meteorology Institute
ENACT	Ensemble of Climate Models
ENSO	El Niño–Southern Oscillation
ESGF	Earth System Grid Federation
GCM	Global Climate Model
GDP	Gross Domestic Product
IOD	Indian Ocean Dipole
IQR	Interquartile Range
ITCZ	Intertropical Convergence Zone
JJAS	June, July, August, and September
MAM	March, April, and May
MK	Mann–Kendall
MME	Multi-Model Ensemble
NMA	National Meteorological Agency
NSE	Nash–Sutcliffe Efficiency
R10mm	Number of Days with Precipitation ≥ 10 mm
R20mm	Number of Days with Precipitation ≥ 20 mm
RCM	Regional Climate Model
RCP	Representative Concentration Pathway
RCP4.5	Representative Concentration Pathway 4.5
RCP8.5	Representative Concentration Pathway 8.5
RMSE	Root Mean Square Error
Rx1day	Maximum 1-day Precipitation
Rx5day	Maximum 5-day Precipitation
SSP	Shared Socioeconomic Pathway
SSP1-2.6	Shared Socioeconomic Pathway 1-2.6
SSP2-4.5	Shared Socioeconomic Pathway 2-4.5
SSP5-8.5	Shared Socioeconomic Pathway 5-8.5

STWJ	Subtropical Westerly Jet
WCRP	World Climate Research Programme
ZMK	Z-statistic of the Mann–Kendall Test

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Authors Contributions

Dejen Terefe Cherinet: Conceptualization, Data curation, Formal Analysis, Methodology, Writing – original draft, Writing – review & editing

Data Availability Statement

The Author obtained observed rainfall data from Ethiopia meteorology institute (EMI), whereas Future rainfall data of CMIP6 /GCMs data were provided by Copernicus climate change service data store for this study.

Conflicts of Interest

The author declares there is no conflict.

Appendix

Table 2. CMIP6 GCM Model for This Study.

GCMs	RF			Tmax			Tmin		
	RMSE	NSE	r	RMSE	NSE	r	RMSE	NSE	r
MPI-ESM1-LR	7.4	-0.1	0.4	8.4	-0.1	0.6	8.9	-9.1	0.7
BCC-CSM2-HR	8.6	-0.5	0.5	7.3	-0.6	0.7	7.2	-5.7	0.5
ACCESS-CM2	5	0.7	0.6	4.9	0.4	0.6	9.2	-10	0.4
CESM2	8.7	-0.5	0.5	6.7	-0.5	0.5	9.6	-11.2	0.6
CNAR-CM6-1	6.1	-5.1	0.3	6.2	-3.3	0.5	2.5	-1.4	0.9
MPI-ESM1-HR	7.7	-0.2	0.6	5.7	-0.2	0.5	6.7	-4.6	0.4
INM-CM5-0	7.8	-0.2	0.6	6.8	-2.2	0.4	8.2	-7.6	0.3
HadGEM3-GC31-LL	7.7	-6.7	0.8	6.3	-5.7	0.6	6.1	-3.8	0.6
Multi model Ensemble	4.6	0	0.9	4.2	0	0.8	2.6	-3.3	0.6

Table 3. Mann Kendall trend test of (RF) under the SSP2-4.5.

Stations	Seasons	ZMK	Sen's slope	P-value	Test interpretation
Adaba	Bega	1.80	0.47	0.07	None significantly increasing
	Belg	2.06	0.79	0.04	Significantly increasing
	Kermit	2.28	0.99	0.02	Significantly increasing
	Annual	3.73	2.01	0.00	Significantly increasing
Adele	Bega	2.95	0.72	0.00	Significantly increasing
	Belg	1.96	1.11	0.05	Significantly increasing

Stations	Seasons	ZMK	Sen's slope	P-value	Test interpretation
Arsi Robe	Kermit	1.56	0.68	0.12	None significantly increasing
	Annual	2.94	2.33	0.00	Significantly increasing
	Bega	1.78	0.71	0.08	None significantly increasing
	Belg	0.11	0.04	0.91	None significantly increasing
	Kiremit	2.44	1.56	0.01	Significantly increasing
	Annual	3.85	2.01	0.00	Significantly increasing
Chole	Bega	3.31	1.45	0.00	Significantly increasing
	Belg	0.67	0.38	0.50	None significantly increasing
	Kiremit	2.79	1.68	0.01	Significantly increasing
	Annual	4.87	3.43	0.00	Significantly increasing
Delosebro	Bega	3.97	1.63	0.00	Significantly increasing
	Belg	0.89	0.46	0.38	None significantly increasing
	Kiremit	4.18	2.18	0.00	Significantly increasing
	Annual	5.17	3.91	0.00	Significantly increasing
Dinisho	Bega	3.64	1.16	0.00	Significantly increasing
	Belg	1.68	0.71	0.09	None significantly increasing
	Kiremit	3.54	0.98	0.00	Significantly increasing
	Annual	4.68	2.79	0.00	Significantly increasing
Dixis	Bega	3.26	1.09	0.00	Significantly increasing
	Belg	0.54	0.26	0.59	None significantly increasing
	Kermit	3.18	2.36	0.00	Significantly increasing
	Annual	4.60	3.32	0.00	Significantly increasing
Dodola	Bega	2.11	1.21	0.03	Significantly increasing
	Belg	-0.33	-0.20	0.74	Non-significant decreasing
	Kermit	2.80	1.83	0.01	Significantly increasing
	Annual	3.17	2.31	0.00	Significantly increasing
Hunte	Bega	2.43	1.26	0.02	Significantly increasing
	Belg	2.42	0.85	0.02	Significantly increasing
	Kiremit	2.95	1.08	0.00	Significantly increasing
	Annual	4.23	3.46	0.00	Significantly increasing
Kofele	Bega	3.04	1.52	0.00	Significantly increasing
	Belg	1.28	0.75	0.20	Non-significant increasing
	Kermit	3.58	1.79	0.00	Significantly increasing
	Annual	5.21	4.01	0.00	Significantly increasing
Meraro	Bega	4.46	1.20	0.00	Significantly increasing
	Belg	2.17	0.80	0.03	Significantly increasing
	Kermit	3.28	1.25	0.00	Significantly increasing
	Annual	5.31	3.37	0.00	Significantly increasing
Jara	Bega	3.12	1.26	0.00	Significantly increasing

Stations	Seasons	ZMK	Sen's slope	P-value	Test interpretation
Seru	Belg	-0.04	-0.05	0.96	Non-significantly decreasing
	Kermit	4.11	1.88	0.00	Significantly increasing
	Annual	4.05	2.92	0.00	Significantly increasing
	Bega	2.16	0.97	0.03	Significantly increasing
	Belg	2.28	1.65	0.02	Significantly increasing
	Kermit	1.80	0.78	0.07	Non-significantly increasing
	Annual	3.13	3.19	0.00	Significantly increasing
	Bega	1.91	0.54	0.06	Non-significant increasing
Silitana	Belg	1.26	0.59	0.21	Non-significant increasing
	Kermit	2.66	1.31	0.01	Significantly increasing
	Annual	3.45	2.27	0.00	Significantly increasing

Table 4. Mann Kendall trend test of (RF) under SSP5-8.5.

Stations	Seasons	ZMK	Sen's slope	P-value	Test interpretation
Adaba	Bega	3.82	1.26	0.00	Significantly increasing
	Belg	3.45	1.36	0.00	Significantly increasing
	Kermit	0.41	0.34	0.68	None significantly increasing
	Annual	4.05	2.99	0.00	Significantly increase
Adele	Bega	3.74	1.45	0.00	Significantly increasing
	Belg	2.74	1.23	0.01	Significantly increasing
	Kermit	2.37	1.54	0.02	Significantly increasing
	Annual	2.94	2.33	0.00	Significantly increasing
Arsi robe	Bega	3.35	1.26	0.00	Significantly increasing
	Belg	1.86	0.56	0.06	Non-significantly increasing
	Kermit	4.23	2.20	0.00	Significantly increasing
	Annual	0.13	3.91	0.00	Significantly increasing
Chole	Bega	3.90	1.71	0.00	Significantly increasing
	Belg	1.80	0.69	0.07	Non-significantly increasing
	Kermit	4.48	2.55	0.00	Significantly increasing
	Annual	0.27	4.82	0.00	Significantly increasing
Delosebro	Bega	4.78	2.09	0.00	Significantly increasing
	Belg	1.56	0.68	0.12	Non-significant increasing
	Kermit	3.96	2.06	0.00	Significantly increasing
	Annual	6.07	4.62	0.00	Significantly increasing
Dinisho	Bega	4.75	2.55	0.00	Significantly increasing
	Belg	2.84	1.37	0.00	Significantly increasing

Stations	Seasons	ZMK	Sen's slope	P-value	Test interpretation
Dixis sude	Kermit	3.67	1.63	0.00	Significantly increasing
	Annual	6.14	5.66	0.00	Significantly increasing
	Bega	3.78	1.64	0.00	Significantly increasing
	Belg	2.11	0.79	0.03	Significantly increasing
	Kermit	4.23	2.57	0.00	Significantly increasing
	Annual	6.40	4.96	0.00	Significantly increasing
Dodola	Bega	4.82	2.10	0.00	Significantly increasing
	Belg	3.28	1.35	0.00	Significantly increasing
	Kermit	3.60	1.71	0.00	Significantly increasing
Hunte	Annual	6.52	5.10	0.00	Significantly increasing
	Bega	4.62	2.70	0.00	Significantly increasing
	Belg	4.13	1.66	0.00	Significantly increasing
	Kermit	3.55	1.26	0.00	Significantly increasing
	Annual	6.13	5.64	0.00	Significantly increasing
Kofele	Bega	3.53	2.59	0.00	Significantly increasing
	Belg	2.52	1.69	0.01	Significantly increasing
	Kermit	4.48	1.37	0.00	Significantly increasing
	Annual	4.89	6.03	0.00	Significantly increasing
	Bega	5.15	1.38	0.00	Significantly increasing
Meraro	Belg	4.39	1.27	0.00	Significantly increasing
	Kermit	3.28	1.10	0.00	Significantly increasing
	Annual	6.47	3.82	0.00	Significantly increasing
	Bega	4.76	2.04	0.00	Significantly increasing
Jara	Belg	1.49	0.57	0.14	Non-significant increasing
	Kermit	3.96	2.02	0.00	Significantly increasing
	Annual	6.09	4.45	0.00	Significantly increasing
Seru	Bega	4.52	3.24	0.00	Significantly increasing
	Belg	1.35	1.14	0.18	Non-significantly decreasing
	Kermit	4.27	3.12	0.00	Significantly increasing
	Annual	5.62	7.10	0.00	Significantly increasing
Silitana	Bega	4.73	1.80	0.00	Significantly increasing
	Belg	3.72	1.47	0.00	Significantly increasing
	Kermit	4.90	2.62	0.00	Significantly increasing
	Annual	6.78	5.56	0.00	Significantly increasing

Table 5. Change of Future (RF) under SSP2-4.5 and SSP5-8.5 Scenarios.

Seasons		SSP2-4.5		SSP5-8.5	
		Period			
		2031-2060	2061-2090	2031-2060	2061-2090
Adaba	Belg	0.2	6.6	14	19.2
	Kermit	0.2	6.6	5	10.9
	Annual	6.6	12.8	5	22.5
Adele	Belg	0.4	16.1	-8.1	7.5
	Kermit	5.1	5.9	27.8	35.1
	Annual	6.4	14.2	17.5	30.3
Arsi robe	Belg	-0.6	-5.8	10.3	10.1
	Kermit	14.4	22.7	22	34.8
	Annual	11.6	17.2	21.1	32.3
Chole	Belg	11.1	7.4	29.2	28.6
	Kermit	-7.2	0.7	23.6	38.2
	Annual	1.3	9.2	32.6	46.4
Delosebro	Belg	9.4	5.8	1.8	1.3
	Kermit	19.1	34.6	7	21.8
	Annual	18.5	28	10.1	22.2
Dinisho	Belg	-27	-23.9	8.7	15.1
	Kermit	-26.5	-21.7	9.9	18.8
	Annual	-22.7	-17.3	16.5	28
Dixis	Belg	24.2	21.7	52.8	84.7
	Kermit	12.9	21.9	29.5	30.3
	Annual	19.1	27.5	22.9	35.7
Dodola	Belg	33.9	16.6	44.2	53.6
	Kermit	-1	10.2	2.6	11.7
	Annual	17.5	22.9	25	39.7
Gasara	Belg	9	10.8	33.3	45.2
	Kermit	6.2	13.3	14	23.9
	Annual	12.5	17.3	29.8	44.8
Hunte	Belg	46.2	60.1	45.3	66.7
	Kermit	-4.3	1	-1.9	5.1
	Annual	29.6	41.3	31.1	49.3
Jara	Belg	-9.4	-16.1	1	-0.2
	Kermit	-19.3	-6.1	-2.8	12.3
	Annual	-7.5	-1.2	4.8	16.7
Kofele	Belg	8.9	9.5	14.3	21.7
	Kermit	13	22.4	-16.4	-9.6

		SSP2-4.5		SSP5-8.5	
Seasons		Period			
		2031-2060	2061-2090	2031-2060	2061-2090
Meraro	Annual	20.6	28.3	12.4	23.5
	Belg	9.4	17	8.3	21.5
	Kermit	0.8	8.5	-0.5	6.4
	Annual	9	19.2	8.9	20.7
Siitana	Belg	-10.4	-10.7	-10.1	-2.8
	Kermit	4.7	10.3	10.9	21.5
	Annual	1.2	4.8	5.2	17.4

Table 6. Change of future T (max) of SSP2-4.5 and SSP5-8.5 scenarios.

		SSP 245		SSP5-8.5	
		Period		Period	
		2031-2060	2061-2090	2031-2060	2061-2090
Adaba	Bega	0.2	1.2	1.5	3.6
	Belg	0.4	1.4	1.2	3.7
	Kirmit	0.2	1.1	1.2	3.5
	Annual	0.3	1.2	1.3	3.6
Arsi robe	Bega	0.5	1.0	3.2	2.3
	Belg	0.6	1.4	3.3	2.7
	Kirmit	0.2	1.1	3.0	2.8
	Annual	0.4	1.2	3.2	2.6
Chole	Bega	0.1	0.6	0.4	1.6
	Belg	0.5	1.4	0.1	2.3
	Kirmit	1.0	1.9	0.9	3.2
	Annual	0.5	1.3	0.5	2.4
Dinisho	Bega	-2.6	-2.1	0.3	1.7
	Belg	-2.8	-2.0	-0.4	1.8
	Kirmit	0.8	1.7	0.8	3.0
	Annual	-1.5	-0.8	0.2	2.2
Hunte	Bega	0.9	1.4	1.7	2.6
	Belg	-0.5	0.2	-0.3	1.5
	Kirmit	-1.2	-0.3	0.0	1.8
	Annual	-0.3	0.4	0.5	2.0
Kofele	Bega	0.6	1.1	4.4	6.3

		SSP 245		SSP5-8.5	
		Period		Period	
		2031-2060	2061-2090	2031-2060	2061-2090
Meraro	Belg	0.9	1.7	4.3	6.7
	Kirmit	1.1	2.0	4.1	6.0
	Annual	0.9	1.6	4.3	6.4
	Bega	1.6	2.1	4.3	3.5
	Belg	1.7	2.5	4.6	4.1
	Kirmit	1.9	2.8	5.0	4.9
	Annual	1.7	2.5	4.6	4.2
	Bega	-0.8	-0.4	0.3	1.0
Seru	Belg	-1.0	-0.2	-0.5	1.2
	Kirmit	-0.8	0.1	0.1	1.9
	Annual	-0.9	-0.1	-0.1	1.4
	Bega	1.7	1.4	1.6	3.0
Jara	Belg	1.8	1.7	1.0	3.1
	Kirmit	2.2	2.4	1.8	4.1
	Annual	1.9	1.8	1.5	3.4

Table 6. Change of future *T* (min) of SSP2-4.5 and SSP5-8.5.

		SSP 2-4.5		SSP5-8.5	
		Period		Period	
		2031-2060	2061-2090	2031-2060	2061-2090
Adaba	Bega	3.1	2.9	0.8	4.8
	Belg	3.2	3.1	1.8	5.7
	Kermit	3.1	3	4.1	5.8
	Annual	1.9	3	2.2	5.2
Adele	Bega	2.1	3.1	3.3	3.8
	Belg	1.8	3.1	3.6	5.3
	Kermit	1.6	2.8	5.9	4.9
	Annual	1.8	3	4.3	5.3
Arsi robe	Bega	2.2	3.2	2.9	5.1
	Belg	2.2	3.5	3.4	4.7
	Kermit	5	4.2	5.5	4.2
	Annual	2	3.2	3.8	4.9
Dinisho	Bega	1.3	2.2	3.3	4.3

		SSP 2-4.5		SSP5-8.5	
		Period		Period	
		2031-2060	2061-2090	2031-2060	2061-2090
	Belg	0.8	2.1	0.4	3.5
	Kermit	1	2.1	1	4
	Annual	1	2.1	5.2	5.2
	Bega	1.3	2.2	0.3	3.4
Hunte	Belg	1.4	2.6	1.3	4.2
	Kermit	1.9	2.9	4	4
	Annual	1.5	2.6	1.9	4.9
	Bega	3.7	4.7	0.7	5
Kofele	Belg	4.2	0.3	1.5	4.1
	Kermit	3.2	4.2	2.9	3.8
	Annual	3.7	4.7	1.7	5.8
	Bega	2.2	3.1	1.9	5
Meraro	Belg	1.8	3	4.1	5
	Kermit	1.8	2.8	4.1	5.2
	Annual	1.9	3	2.7	5.7
	Bega	2	3	3.9	4.7
Seru	Belg	1.7	3	4.4	5.5
	Kermit	1.6	2.8	4.1	4.8
	Annual	1.8	3	5.1	4.8
	Bega	1.6	2.6	4.3	5.4
Jara	Belg	1.1	2.5	4.3	4.2
	Kermit	1.1	2.3	4.3	4
	Annual	1.3	2.5	4.3	5.7

References

- [1] H. Lee *et al.*, "IPCC, 2023: Climate Change 2023: Synthesis Report, Summary for Policymakers. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)] <https://doi.org/10.59327/ipcc/ar6-9789291691647.003>
- [2] I. C. Change, "impacts, adaptation and vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change," *O Roberts DC Tignor M Poloczanska ES Mintenbeck K Alegr á Craig M Langsdorf L öschke Möller V Al Eds*, vol. 3056, 2022. <https://doi.org/10.1017/9781009325844>
- [3] T. Gashaw *et al.*, "Local and regional climate trends and variabilities in Ethiopia: Implications for climate change adaptations," *Environ. Chall.*, vol. 13, p. 100794, 2023. <https://doi.org/10.1016/j.envc.2023.100794>
- [4] V. Masson-Delmotte *et al.*, "Climate change 2021: the physical science basis," *Contrib. Work. Group Sixth Assess. Rep. Intergov. Panel Clim. Change*, vol. 2, no. 1, p. 2391, 2021. <https://doi.org/10.1017/9781009157896>
- [5] J. Kennedy *et al.*, "State of the Climate 2024. Update for COP29," 2024. <https://doi.org/10.54377/ddf6-a24c>
- [6] N. C. Emiru *et al.*, "Impact of climate change on the hydrology of the Upper Awash River Basin, Ethiopia," *Hydrology*, vol. 9, no. 1, p. 3, 2021. <https://doi.org/10.3390/hydrology9010003>

- [7] S. Beshir, A. Moges, and M. Dananto, "Climate change projections using CMIP6 GCMs and downscaling approaches in the Upper Wabe Shebele Basin, Ethiopia," *Sci. Rep.*, vol. 15, no. 1, p. 39521, 2025. <https://doi.org/10.1038/s41598-025-23194-6>
- [8] L. A. Espinosa, M. M. Portela, J. P. Matos, and S. Gharbia, "Climate change trends in a European coastal metropolitan area: Rainfall, temperature, and extreme events (1864–2021)," *Atmosphere*, vol. 13, no. 12, p. 1995, 2022. <https://doi.org/10.3390/atmos13121995>
- [9] S. Kone, A. Balde, P. Zahonogo, and S. Sanfo, "A systematic review of recent estimations of climate change impact on agriculture and adaptation strategies perspectives in Africa," *Mitig. Adapt. Strateg. Glob. Change*, vol. 29, no. 2, p. 18, 2024. <https://doi.org/10.1007/s11027-024-10115-7>
- [10] Y. Di, X. Wei, and X. Yin-long, "A review on impacts of climate change on water resource and agriculture in Africa," *Chin. J. Agrometeorol.*, vol. 37, no. 03, p. 259, 2016. <https://doi.org/10.1596/1813-9450-4290>
- [11] K. Ngongolo and L. Gayo, "Climate change impacts and mitigation strategies in Sub-Saharan Africa's livestock production sector. A brief review," *Pastor. Res. Policy Pract.*, vol. 15, p. 14225, 2025. <https://doi.org/10.3389/past.2025.14225>
- [12] M. K. Adamu and K. D. Negeso, "Effect of climate change on agricultural output in Ethiopia," *J. Econ. Dev. Environ. People*, vol. 9, no. 3, p. 6, 2020. <https://doi.org/10.26458/jedep.v9i3.665>
- [13] T. T. Deressa, R. M. Hassan, and C. Ringler, "Perception of and adaptation to climate change by farmers in the Nile basin of Ethiopia," *J. Agric. Sci.*, vol. 149, no. 1, pp. 23–31, 2011. <https://doi.org/10.1017/s0021859610000687>
- [14] M. Asfew and A. Bedemo, "Impact of climate change on cereal crops production in Ethiopia," *Adv. Agric.*, vol. 2022, no. 1, p. 2208694, 2022. <https://doi.org/10.1155/2022/2208694>
- [15] U. Kim and J. J. Kaluarachchi, "Climate change impacts on water resources in the upper blue Nile River Basin, Ethiopia 1," *JAWRA J. Am. Water Resour. Assoc.*, vol. 45, no. 6, pp. 1361–1378, 2009. <https://doi.org/10.1111/j.1752-1688.2009.00369.x>
- [16] T. T. Gule, B. T. Hailu, and B. Lemma, "The ripple effect of climate change: Assessing the impacts on water quality and hydrology in Addis Ababa City (Akaki Catchment)," *Scientifica*, vol. 2024, no. 1, p. 8824622, 2024. <https://doi.org/10.1155/2024/8824622>
- [17] A. Bezu, "Analyzing impacts of climate variability and changes in Ethiopia: A review," *Am. J. Mod. Energy*, vol. 6, no. 3, pp. 65–76, 2020. <https://doi.org/10.11648/j.ajme.20200603.11>
- [18] A. T. Toni, A. Malcherek, and A. K. Kassa, "Agroclimatic zone-based analysis of rainfall variability and trends in the Wabi Shebele River Basin, Ethiopia," *Water*, vol. 14, no. 22, p. 3699, 2022. <https://doi.org/10.3390/w14223699>
- [19] V. Eyring *et al.*, "Overview of the Coupled Model Intercomparison Project Phase 6 (CMIP6) experimental design and organization," *Geosci. Model Dev.*, vol. 9, no. 5, pp. 1937–1958, 2016. <https://doi.org/10.5194/gmd-9-1937-2016>
- [20] M. Almazroui *et al.*, "Projected Change in Temperature and Precipitation Over Africa from CMIP6: M. Almazroui *et al.*," *Earth Syst. Environ.*, vol. 4, no. 3, pp. 455–475, 2020. <https://doi.org/10.1007/s41748-020-00161-x>
- [21] A. G. Mengistu, W. A. Tesfuhoney, Y. E. Woyessa, A. A. Ejigu, and M. D. Alemu, "Future Climate and Hydrological Extremes in Ethiopia: A CMIP6-Based Analysis," *Adv. Meteorol.*, vol. 2025, no. 1, p. 5571487, 2025. <https://doi.org/10.1155/adme/5571487>
- [22] F. M. Rettie, S. Gayler, T. K. Weber, K. Tesfaye, and T. Streck, "High-resolution CMIP6 climate projections for Ethiopia using the gridded statistical downscaling method," *Sci. Data*, vol. 10, no. 1, p. 442, 2023. <https://doi.org/10.1038/s41597-023-02337-2>
- [23] M. A. Gurara, N. B. Jilo, and A. D. Tolche, "Modelling climate change impact on the streamflow in the Upper Wabe Bridge watershed in Wabe Shebele River Basin, Ethiopia," *Int. J. River Basin Manag.*, vol. 21, no. 2, pp. 181–193, 2023. <https://doi.org/10.1080/15715124.2021.1935978>
- [24] A. Belay *et al.*, "Analysis of climate variability and trends in Southern Ethiopia," *Climate*, vol. 9, no. 6, p. 96, 2021. <https://doi.org/10.3390/cli9060096>
- [25] T. K. Beyene, M. K. Jain, B. K. Yadav, and A. Agarwal, "Multiscale investigation of precipitation extremes over Ethiopia and teleconnections to large-scale climate anomalies," *Stoch. Environ. Res. Risk Assess.*, vol. 36, no. 5, pp. 1503–1519, 2022. <https://doi.org/10.1007/s00477-021-02120-y>
- [26] G. T. Diro, D. Grimes, and E. Black, "Large scale features affecting Ethiopian rainfall," in *African climate and climate change: physical, social and political perspectives*, Springer, 2011, pp. 13–50. https://doi.org/10.1007/978-90-481-3842-5_2
- [27] T. Reek, S. R. Doty, and T. W. Owen, "A deterministic approach to the validation of historical daily temperature and precipitation data from the cooperative network," *Bull. Am. Meteorol. Soc.*, vol. 73, no. 6, pp. 753–765, 1992. [https://doi.org/10.1175/1520-0477\(1992\)073%3C0753](https://doi.org/10.1175/1520-0477(1992)073%3C0753)
- [28] J. W. Tukey, *Exploratory data analysis*, vol. 2. Reading, MA, 1977. <https://doi.org/10.2307/2529486>
- [29] E. P. Diaconescu, P. Gachon, J. Scinocca, and R. Laprise, "Evaluation of daily precipitation statistics and monsoon onset/retreat over western Sahel in multiple data sets," *Clim. Dyn.*, vol. 45, no. 5, pp. 1325–1354, 2015. <https://doi.org/10.1007/s00382-014-2383-2>
- [30] J. E. Walsh, W. L. Chapman, V. Romanovsky, J. H. Christensen, and M. Stendel, "Global climate model performance over Alaska and Greenland," *J. Clim.*, vol. 21, no. 23, pp. 6156–6174, 2008. <https://doi.org/10.1175/2008jcli2163.1>
- [31] C. J. Willmott and K. Matsuura, "Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance," *Clim. Res.*, vol. 30, pp. 79–82, 2005. <https://doi.org/10.3354/cr030079>

- [32] S. Shrestha, R. Kafle, and V. P. Pandey, "Evaluation of index-overlay methods for groundwater vulnerability and risk assessment in Kathmandu Valley, Nepal," *Sci. Total Environ.*, vol. 575, pp. 779–790, 2017. <https://doi.org/10.1016/j.scitotenv.2016.09.141>
- [33] S. A. Shah and M. Kiran, "Mann-Kendall test: trend analysis of temperature, rainfall and discharge of Ghotki feeder canal in district Ghotki, Sindh, Pakistan," *Environ. Ecosyst. Sci. EES*, vol. 5, no. 2, pp. 137–142, 2021. <https://doi.org/10.26480/ees.02.2021.137.142>
- [34] A. A. Alaminie, S. A. Tilahun, S. A. Legesse, F. A. Zimale, G. B. Tarkegn, and M. R. Jury, "Evaluation of past and future climate trends under CMIP6 scenarios for the UBNB (Abay), Ethiopia," *Water*, vol. 13, no. 15, p. 2110, 2021. <https://doi.org/10.3390/w13152110>
- [35] S. B. Atanaw, F. A. Zimale, T. Ayenew, and G. T. Ayele, "Modeling future hydrological responses through parameter optimization and climate change scenarios in Dirima Watershed, Ethiopia," *Model. Earth Syst. Environ.*, pp. 1–19, 2023. <https://doi.org/10.1007/s40808-023-01817-z>
- [36] T. A. Feyissa, T. A. Demissie, F. Saathoff, and A. Gebissa, "Evaluation of General Circulation Models CMIP6 Performance and Future Climate Change over the Omo River Basin, Ethiopia," *Sustainability*, vol. 15, no. 8, p. 6507, 2023. <https://doi.org/10.3390/su15086507>
- [37] S. H. Gebresellase, Z. Wu, H. Xu, and W. I. Muhammad, "Evaluation and selection of CMIP6 climate models in Upper Awash Basin (UBA), Ethiopia: Evaluation and selection of CMIP6 climate models in Upper Awash Basin (UBA), Ethiopia," *Theor. Appl. Climatol.*, vol. 149, no. 3–4, pp. 1521–1547, 2022. <https://doi.org/10.1007/s00704-022-04056-x>
- [38] M. Seker and V. Gumus, "Projection of temperature and precipitation in the Mediterranean region through multi-model ensemble from CMIP6," *Atmos. Res.*, vol. 280, p. 106440, 2022. <https://doi.org/10.1016/j.atmosres.2022.106440>
- [39] B. G. Gobie, A. D. Asamnew, and B. A. Habtemichael, "Projected annual precipitation trend in Ethiopia under CMIP6 models in the 21st century," *Model. Earth Syst. Environ.*, vol. 10, no. 3, pp. 3633–3644, 2024. <https://doi.org/10.1007/s40808-024-01971-y>
- [40] D. Berhanu, T. Alamirew, M. T. Taye, D. Tibebe, S. Gebrehiwot, and G. Zeleke, "Evaluation of CMIP6 models in reproducing observed rainfall over Ethiopia," *J. Water Clim. Change*, vol. 14, no. 8, pp. 2583–2605, 2023. <https://doi.org/10.2166/wcc.2023.502>
- [41] A. B. Tagele *et al.*, "Evaluating the performance of selected CMIP6 GCMs for simulations of historical temperature over Ethiopia," *Theor. Appl. Climatol.*, vol. 157, no. 2, p. 121, 2026. <https://doi.org/10.1007/s00704-026-06066-5>
- [42] D. Maraun, "Bias correcting climate change simulations-a critical review," *Curr. Clim. Change Rep.*, vol. 2, no. 4, pp. 211–220, 2016. <https://doi.org/10.1007/s40641-016-0050-x>
- [43] C. Teutschbein and J. Seibert, "Bias correction of regional climate model simulations for hydrological climate-change impact studies: Review and evaluation of different methods," *J. Hydrol.*, vol. 456, pp. 12–29, 2012. <https://doi.org/10.1016/j.jhydrol.2012.05.052>
- [44] M. H. Kebede, A. M. Ahmed, D. A. Birhan, G. A. Damot, and S. A. Legesse, "Projections of precipitation and temperature changes and trends using CMIP6 global climate models in the Eastern Amhara, Northeastern, Ethiopia," *Meteorol. Appl.*, vol. 33, no. 1, p. e70145, 2026. <https://doi.org/10.1002/met.70145>
- [45] X. Wang, C. Dai, G. Liu, X. Yang, J. Jing, and Q. Ru, "Runoff prediction in the songhua river basin based on wep model," *Hydrology*, vol. 12, no. 10, p. 266, 2025. <https://doi.org/10.3390/hydrology12100266>
- [46] S. Gebrechorkos, M. T. Taye, B. Birhanu, D. Solomon, and T. Demissie, "Future changes in climate and hydroclimate extremes in East Africa," *Earths Future*, vol. 11, no. 2, p. e2022EF003011, 2023. <https://doi.org/10.1029/2022ef003011>
- [47] A. Asfaw, B. Simane, A. Hassen, and A. Bantider, "Variability and time series trend analysis of rainfall and temperature in northcentral Ethiopia: A case study in Woleka sub-basin," *Weather Clim. Extrem.*, vol. 19, pp. 29–41, 2018. <https://doi.org/10.1016/j.wace.2017.12.002>
- [48] S. Admassu and A. H. Seid, "Analysis of rainfall trend in Ethiopia," *Ethiop. J. Sci. Technol.*, vol. 3, no. 2, pp. 15–30, 2006.
- [49] P. Iradukunda, E. M. Mwanaumo, and J. Kabika, "Hydroclimatic trend analysis and projection in Africa tropical urban regions: Cases of Lusaka, Zambia and Kigali, Rwanda," *Urban Clim.*, vol. 52, p. 101680, 2023. <https://doi.org/10.1016/j.uclim.2023.101680>
- [50] K. N. Disasa *et al.*, "Trend Analysis and Projection of Climate Variables Using the LARS-WG Model in Coastal Areas of China," 2024. <https://doi.org/10.21203/rs.3.rs-4015632/v1>
- [51] Y. A. Balcha, A. Malcherek, and T. Alamirew, "Understanding future climate in the upper awash basin (UASB) with selected climate model outputs under CMIP6," *Climate*, vol. 10, no. 12, p. 185, 2022. <https://doi.org/10.3390/cli10120185>
- [52] D. Banerjee and C. Singh, "An appraisal of seasonal precipitation dynamics over the North-West Himalayan region under future warming scenarios," *Int. J. Climatol.*, vol. 42, no. 4, pp. 2328–2350, 2022. <https://doi.org/10.1002/joc.7368>
- [53] X. Guan, J. Zhang, Z. Bao, C. Liu, J. Jin, and G. Wang, "Past variations and future projection of runoff in typical basins in 10 water zones, China," *Sci. Total Environ.*, vol. 798, p. 149277, 2021. <https://doi.org/10.1016/j.scitotenv.2021.149277>
- [54] K. Haleem, A. U. Khan, J. Khan, A. A. Ghanim, and A. M. Al-Areeq, "Evaluating future streamflow patterns under SSP245 scenarios: insights from CMIP6," *Sustainability*, vol. 15, no. 22, p. 16117, 2023. <https://doi.org/10.3390/su152216117>
- [55] K. Karan, D. Singh, P. K. Singh, B. Bharati, T. P. Singh, and R. Berndtsson, "Implications of future climate change on crop and irrigation water requirements in a semi-arid river basin using CMIP6 GCMs," *J. Arid Land*, vol. 14, no. 11, pp. 1234–1257, 2022. <https://doi.org/10.1007/s40333-022-0081-1>

- [56] A. T. Tadase and A. K. Tekile, "Future projections of climate variables and meteorological drought: insight from CMIP6 models in Southeast Ethiopia," *Dyn. Atmos. Oceans*, vol. 110, p. 101538, 2025.
<https://doi.org/10.1016/j.dynatmoce.2025.101538>
- [57] Q. Wang *et al.*, "Exploring future trends of precipitation and runoff in arid regions under different scenarios based on a bias-corrected CMIP6 model," *J. Hydrol.*, vol. 630, p. 130666, 2024.
<https://doi.org/10.1016/j.jhydrol.2024.130666>
- [58] Y. Yue, D. Yan, Q. Yue, G. Ji, and Z. Wang, "Future changes in precipitation and temperature over the Yangtze River Basin in China based on CMIP6 GCMs," *Atmos. Res.*, vol. 264, p. 105828, 2021.
<https://doi.org/10.1016/j.atmosres.2021.105828>
- [59] N. M. Reddy and S. Saravanan, "Extreme precipitation indices over India using CMIP6: a special emphasis on the SSP585 scenario," *Environ. Sci. Pollut. Res.*, vol. 30, no. 16, pp. 47119–47143, 2023.
<https://doi.org/10.1007/s11356-023-25649-7>
- [60] M. Shafeeque and Y. Luo, "A multi-perspective approach for selecting CMIP6 scenarios to project climate change impacts on glacio-hydrology with a case study in Upper Indus river basin," *J. Hydrol.*, vol. 599, p. 126466, 2021.
<https://doi.org/10.1016/j.jhydrol.2021.126466>
- [61] N. A. Hersi, D. M. Mulungu, and J. Nobert, "Prediction of future climate in semi-arid catchment under CMIP6 scenarios: A case study of Bahi (Manyoni) catchment in Internal Drainage basin (IDB), Tanzania," *Phys. Chem. Earth Parts ABC*, vol. 129, p. 103309, 2023.
<https://doi.org/10.1016/j.pce.2022.103309>
- [62] C. Assessment, "Fourth national climate assessment," 2018.
<https://doi.org/10.7930/nca4.2018.ap2>
- [63] Q. You, Z. Cai, F. Wu, Z. Jiang, N. Pepin, and S. S. Shen, "Temperature dataset of CMIP6 models over China: evaluation, trend and uncertainty," *Clim. Dyn.*, vol. 57, pp. 17–35, 2021.
<https://doi.org/10.1007/s00382-021-05691-2>
- [64] P. Das *et al.*, "Historical and projected changes in Extreme High Temperature events over East Africa and associated with meteorological conditions using CMIP6 models," *Glob. Planet. Change*, vol. 222, p. 104068, 2023.
<https://doi.org/10.1016/j.gloplacha.2023.104068>
- [65] S. Kibebew, N. Dechassa, Y. Alemayehu, F. Merga, and G. Megersa, "Climate trends, variability, and changes in extreme indices under future projections over Central Oromia, Ethiopia," *Discov. Environ.*, vol. 3, no. 1, p. 271, 2025.
<https://doi.org/10.1007/s44274-025-00476-4>
- [66] F. Majdi, S. A. Hosseini, A. Karbalaee, M. Kaseri, and S. Marjanian, "Future projection of precipitation and temperature changes in the Middle East and North Africa (MENA) region based on CMIP6," *Theor. Appl. Climatol.*, pp. 1–14, 2022.
<https://doi.org/10.1007/s00704-021-03916-2>
- [67] D. Carvalho, S. Cardoso Pereira, and A. Rocha, "Future surface temperatures over Europe according to CMIP6 climate projections: an analysis with original and bias-corrected data," *Clim. Change*, vol. 167, pp. 1–17, 2021.
<https://doi.org/10.1007/s10584-021-03159-0>
- [68] B. Ayugi, V. Dike, H. Ngoma, H. Babaousmail, R. Mumo, and V. Ongoma, "Future changes in precipitation extremes over East Africa based on CMIP6 models," *Water*, vol. 13, no. 17, p. 2358, 2021.
<https://doi.org/10.3390/w13172358>
- [69] S. B. Ajjur and S. G. Al-Ghamdi, "Global hotspots for future absolute temperature extremes from CMIP6 models," *Earth Space Sci.*, vol. 8, no. 9, p. e2021EA001817, 2021.
<https://doi.org/10.1029/2021ea001817>
- [70] Y.-W. Choi, D. J. Campbell, and E. A. Eltahir, "Near-term regional climate change in East Africa," *Clim. Dyn.*, vol. 61, no. 1, pp. 961–978, 2023.
<https://doi.org/10.1007/s00382-022-06591-9>
- [71] X. Fan, Q. Duan, C. Shen, Y. Wu, and C. Xing, "Global surface air temperatures in CMIP6: historical performance and future changes," *Environ. Res. Lett.*, vol. 15, no. 10, p. 104056, 2020.
<https://doi.org/10.1088/1748-9326/abb051>
- [72] M. G. Alemu, M. A. Wubneh, D. Sahlu, and F. A. Zimale, "Spatiotemporal change of climate extremes under the projection of CMIP6 model analysis over Awash Basin, Ethiopia," *Sustain. Water Resour. Manag.*, vol. 9, no. 6, p. 195, 2023.
<https://doi.org/10.1007/s40899-023-00981-5>
- [73] E. Meskelu, M. Ayana, and D. Birhanu, "Assessment of climate change impacts on crop and irrigation water demand in the Awash River basin of Ethiopia using CMIP6 models," *Discov. Water*, vol. 5, no. 1, p. 112, 2025.
<https://doi.org/10.1007/s43832-025-00307-w>
- [74] X. Fan, Q. Duan, C. Shen, Y. Wu, and C. Xing, "Evaluation of historical CMIP6 model simulations and future projections of temperature over the Pan-Third Pole region," *Environ. Sci. Pollut. Res.*, vol. 29, no. 18, pp. 26214–26229, 2022.
<https://doi.org/10.1007/s11356-021-17474-7>
- [75] T. L. Kumar, B. Vinodhkumar, K. K. Rao, J. Chowdary, K. K. Osuri, and S. Desamsetti, "Insights from the bias-corrected simulations of CMIP6 in India's future climate," *Glob. Planet. Change*, vol. 226, p. 104137, 2023.
<https://doi.org/10.1016/j.gloplacha.2023.104137>
- [76] S. K. Mondal *et al.*, "Projected changes in temperature, precipitation and potential evapotranspiration across Indus River Basin at 1.5–3.0 C warming levels using CMIP6-GCMs," *Sci. Total Environ.*, vol. 789, p. 147867, 2021.
<https://doi.org/10.1016/j.scitotenv.2021.147867>
- [77] B. T. Gebisa, W. T. Dibaba, and A. Kabeta, "Evaluation of historical CMIP6 model simulations and future climate change projections in the Baro River Basin," *J. Water Clim. Change*, vol. 14, no. 8, pp. 2680–2705, 2023.
<https://doi.org/10.2166/wcc.2023.032>
- [78] F. M. Chanie, "Evaluation of CMIP6 model performance and future climate projections over the Genale dawa river basin, Ethiopia," *Sci. Rep.*, vol. 16, no. 1, p. 361, 2025.
<https://doi.org/10.1038/s41598-025-29498-x>